

The Masonry Fireplace — Parts and Materials

Naming the pieces, and the materials question you have missed five times.

Firebox	The chamber the fire burns in, lined with firebrick.
Ash dump	An opening in a fireplace hearth into which ashes may be swept into a pit below.
Corbel	The outward projection of a masonry course beyond the course below it, so masonry can carry something wider than its base.
Lintel	What supports masonry installed above a masonry fireplace opening.
Wythe	A single vertical thickness of brick. Flues in masonry chimneys must be separated by at least one wythe of brick, so a leak from one flue cannot cross into the other.

The numbers here

Firebrick lining inside a firebox

2 in

YOU HAVE MISSED THIS 3 TIMES

An opening in a fireplace hearth into which ashes may be swept into a pit below is called:

An ash dump.

YOU HAVE MISSED THIS 2 TIMES

Flues in masonry chimneys should be separated by at least one ____ of brick.

Wythe.

The crown

YOU HAVE MISSED THIS 5 TIMES

All of the following are improper materials for a masonry chimney crown EXCEPT:

Concrete.

Five times missed, and the EXCEPT is what makes it hard: you are being asked for the one material that IS correct. The crown is the sloped cap on top of the chimney, fully exposed to weather with no protection above it, so it must be cast concrete. Mortar, brick and parge coats all crack and fail there.

The flue liner

The numbers here

Clay flue liner extends above the crown

4 to 6 in

YOU HAVE MISSED THIS 2 TIMES

A clay flue liner should extend _____ inches above the crown.

Four to six.

Refractory mortar	What a fireplace clay flue liner is held in place with. Ordinary mortar cannot survive flue temperatures.
Mitered seams	What a masonry fireplace clay liner needs where it changes direction, so the sections meet cleanly at the angle with no gap for gases to escape through.
Grout	Mortar mixed with extra water so it can be poured into block cores.
Fire-resistant caulk	What a penetration through a firebox wall, such as for a gas line, requires.
Pyrolysis	The process by which heat changes wood chemistry and makes it easier to ignite. It is why long exposure to moderate heat is dangerous, not just brief exposure to high heat.

Say it back

Cover the page. Answer each one out loud.

1. What is an ash dump, a corbel and a lintel?
2. What is a wythe, and why must flues be separated by one?
3. Correct material for a chimney crown, and why the others fail?
4. How far does a clay flue liner extend above the crown?
5. What holds a clay flue liner in place, and what seams does it need at a bend?
6. What is grout, and what does a gas line penetration through a firebox need?
7. Define pyrolysis, and say why it matters.